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Practical R&D Roadmap for Environment-Friendly Thermonuclear Explosions

Micro & macro thermonuke ignition, antimatter-triggered fusion, *plasma-jet DEWs*, *GRASERs*, and ultrafast space propulsion

This document synthesizes four decades of open-literature research on fissile-free thermonuclear ignition — from Gsponer & Hurni's 1986 antimatter-induced fusion paper through NIF's 2022–2025 ignition milestones, the ALPHA November-2025 antihydrogen-production breakthrough, and General Fusion's LM26 magnetized-target results. It maps four application tracks — directed energy, gamma-ray lasers, space propulsion, X-ray-laser pumping — to a TRL roadmap and to the CTBT/NPT non-proliferation regime. No classified or weapon-construction information is contained herein.

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Executive Summary

Author: Ren Timer, CTO — Aumnium Technology (*hardagent.one* · *aumnium.tech/products* · *stoXswap.com/anonhost* · *stoXswap.com/finance*). This technical assessment synthesizes four decades of open, peer-reviewed literature on engineering pathways to **fissile-free thermonuclear explosions** — that is, micro- and macro-scale fusion bursts that do not require a plutonium or highly-enriched-uranium fission primary. The motivating source is Gsponer & Hurni's 1986 paper *The physics of antimatter induced fusion and thermonuclear explosions* (arXiv:physics/0507114, 2005 republication), which first argued at a scientific conference that as few as 10^{18} antiprotons could trigger a 1-kiloton-class thermonuclear device. We update that 1986 analysis with four material developments of the 2022-2026 period:

(1) The U.S. National Ignition Facility (NIF) achieved controlled inertial-confinement fusion ignition in December 2022 (3.15 MJ output, 1.5× target gain) and has since repeated ignition on more than eleven shots, with yields climbing to 5.2 MJ (Feb 2024) and an apparent 8.6 MJ record (April 2025). For the first time, the thermonuclear burn-wave conditions Gsponer & Hurni calculated for an antimatter-catalyzed spark have been reproduced in a non-fission laboratory device.

(2) The ALPHA collaboration at CERN announced (November 18, 2025) a new sympathetic-cooling technique using laser-cooled Be^+ ions that increases antihydrogen production rate by a factor of eight. The BASE collaboration has demonstrated antiproton Penning-trap storage for ~600 days (2024), and BASE-STEP has transported antiprotons by road across the CERN site (Leonhardt et al., Nature 2026). Antimatter handling has moved from a seconds-scale curiosity in 1986 to a road-transportable industrial capability in 2026.

(3) Magnetized Target Fusion (MTF) — the closest-to-compact non-fission, non-antimatter ignition route — has matured rapidly. General Fusion's LM26 machine demonstrated stable magnetized plasmas with confinement times exceeding 10 ms, plasma temperatures near 0.72 keV, and sustained fusion-neutron yields of ~6×10^8 neutrons/s in late 2024, with the first UKAEA-co-authored world-first MTF milestone reached in October 2026. A 1-tonne-class MTF pure-fusion device is now openly described in the technical literature as deliverable with present-day technology.

(4) The arms-control literature has caught up. The Bulletin of the Atomic Scientists (November 2024) argues that the December 2022 NIF shot may technically have been a "nuclear explosion" under the Comprehensive Test Ban Treaty (CTBT) — the first time the treaty's "any other nuclear explosion" clause has arguably been triggered. The IEER's 1998 *Dangerous Thermonuclear Quest* report remains the canonical policy synthesis.

Roadmap scope.

We map four application tracks to a technology-readiness-level (TRL) framework with a 2025-2045 timeline: (A) plasma-jet directed-energy weapons (DEWs) driven by shaped thermonuclear microexplosions; (B) GRASERs — gamma-ray lasers pumped by nuclear-explosion X-ray baths; (C) Antimatter-Initiated Microfusion (AIM) for ultrafast space propulsion; and (D) X-ray laser pumping in the classical Teller-Wood lineage. For each track we tabulate required antiproton count (where applicable), yield/mass envelope, special-material dependencies, and current TRL. We then describe the non-proliferation dimensions: CTBT Article I scope, NPT Article I/II implications, the verification gap created by the absence of fissile material, and the policy options that have been publicly proposed (no-first-pure-fusion-weapon declarations, threshold-yield declarations, antimatter-facility safeguarding).

No classified, weaponization, or build-instruction content is contained in this document. *All technical data are drawn from open scientific literature. The intended audience is policy and defense analysts requiring a technical-understanding brief, not engineers seeking design specifications. Where quantitative estimates appear, they are reproduced from cited open sources and bounded by the assumptions of those sources.*

Chapter 1 Background and Scope

1.1 Historical arc: 1956 to 2026

The idea that antiproton annihilation could be harnessed to initiate thermonuclear burn was first articulated in quantitative form by Duerr and Teller in 1956, who speculated that an antiproton would encounter a ~900 MeV attractive nuclear potential as it approached a nucleus — a finding later refined to the Fermi-motion recoil and pion-nucleus cascade model used today. Edward Teller's career-long interest in antimatter, documented in Morgan & Hughes (1970, p. 1399) and in Teller's own subsequent publications, established the conceptual foundation that Andrei Sakharov independently pursued on the Soviet side (Sakharov, *Collected Scientific Work*, 1982). The antimatter-annihilation route is therefore not a recent invention; it is the third generation of nuclear-explosive physics after fission (1945) and two-stage thermonuclear (1952), and was studied by the principal weapons designers of both major nuclear powers.

Gsponer and Hurni's 1986 paper — delivered at the 4th International Conference on Emerging Nuclear Energy Systems in Madrid and republished on arXiv in 2005 — was the first scientific-conference presentation arguing that fewer than a few micrograms of antimatter could trigger a large-scale thermonuclear explosion. The paper's central quantitative result, that the antiproton count required to start a burn wave in DT or Li_2DT scales as $N \approx 10^{21} / \kappa^2$ (where κ is the fuel compression factor), remains the reference estimate. Their paper also introduced the 1-kt antimatter bomb configuration (Figure 2 of the source PDF) in which one microgram of antihydrogen is levitated in a microcryostat at the centre of a 100 g Li_2DT sphere, imploded by chemical explosives.

Between 1986 and 2022 the field advanced incrementally: the CERN LEAR (Low-Energy Antiproton Ring) ran from 1982 to 1996; ATHENA produced the first cold antihydrogen in 2002; ALPHA trapped antihydrogen for 1000 s in 2011; ELENA came online in 2018 to increase antiproton density. Theorists refined the antiproton-count estimates: Perkins, Orth & Tabak (LLNL, *Nuclear Fusion* 44, 2004) introduced Antimatter-Initiated Microfusion (AIM) for space propulsion; Shmatov (Ioffe Institute, *JBIS* 58, 2005) showed that doping DT fuel with U-238 substantially reduces the antiproton count by leveraging the heavy-nucleus local-deposition term that Gsponer & Hurni had identified. The 2023 paper by Azizi et al. (Physica Scripta 98) extended Shmatov's line to fast-ignition targets.

1.2 The 2022-2026 inflection

What changed in 2022 is that the inertial-confinement-fusion (ICF) community crossed the ignition threshold in a laboratory device. On 5 December 2022, NIF delivered 2.05 MJ of ultraviolet laser energy to a deuterium-tritium target and got 3.15 MJ of fusion energy out — first controlled ignition and first net target gain ever. The shot released the energy of roughly three sticks of dynamite in nanoseconds — the time scale of a thermonuclear burn wave — and it did so without any fissile material. The Bulletin of the Atomic Scientists (November 2024) argues that this shot may technically have been a nuclear explosion under the CTBT's Article I scope, which prohibits "any nuclear weapon test explosion or any other nuclear explosion" regardless of yield or purpose.

Two subsequent developments in 2024-2026 hardened the inflection. First, ALPHA's November 2025 sympathetic-cooling breakthrough raised antihydrogen production eight-fold — 15,000 trapped antihydrogen atoms in seven hours versus the previous ten-week cadence — making antihydrogen accumulation a

near-routine overnight procedure. Second, General Fusion's LM26 magnetized-target-fusion machine reached fusion-relevant plasma conditions ($T > 0.7$ keV, $\tau_E > 10$ ms, sustained neutron emission) on a path explicitly conceived by Sakharov in 1951 and pursued by joint US-Russian teams at Los Alamos / Arzamas-16 in the 1990s. MTF is the only ignition route that scales with chemical-explosive compression — the exact approach used in the implosion stage of every existing fission primary — making it the closest-to-compact fissile-free ignition concept.

1.3 Definitions

Throughout this document we use the following definitions, drawn from the open arms-control literature (notably Jones & von Hippel, *Science & Global Security* 6, 1998, and IEER, *Dangerous Thermonuclear Quest*, 1998):

Term	Definition used in this report
Pure fusion weapon	A thermonuclear explosive that produces its yield entirely from fusion reactions, requiring no fissile material (plutonium or HEU) for ignition.
Fissile-free ignition	Achievement of thermonuclear burn conditions without a fission primary; compatible with civilian research, but also a prerequisite for a pure fusion weapon.
Micro explosion	A nuclear-fusion energy release in the milliton to ~1 ton TNT-equivalent range, comparable to a laboratory ICF shot or a tactical-scale pure-fusion device.
Macro explosion	A nuclear-fusion energy release in the kiloton to megaton TNT-equivalent range, comparable to a strategic-scale thermonuclear weapon.
Antimatter-catalyzed microfusion (ACMF)	A fusion ignition concept in which antiproton annihilation provides the spark energy; the antiprotons are consumed, so "catalyzed" is technically a misnomer (Wikipedia, 2024).
Environment-friendly	Gsponer & Hurni's term for reduced-fission-fusion-fraction thermonuclear devices that produce minimal radioactive fallout relative to a fission primary of comparable yield; a non-proliferation framing, not an environmental claim.
DEW	Directed-energy weapon. In this report: weapons that use a thermonuclear microexplosion as the energy source for a directed plasma jet or a coherent gamma-ray burst.
GRASER	Gamma-ray laser. A hypothetical device in which a nuclear-isomer population inversion is pumped by an X-ray bath (from a thermonuclear microexplosion) and emits a coherent gamma-ray beam. No working device exists.
AIM	Antimatter-Initiated Microfusion: the Penn State / Perkins-Orth-Tabak propulsion concept in which a trapped antiproton cloud is brought into contact with a DT or D-^{3}He pellet; specific energy ~90 MJ/μg antiprotons.

Table 1.1 — Working definitions used throughout this report.

1.4 Why "environment-friendly" framing matters for non-proliferation

Gsponer & Hurni chose the phrase "environment-friendly" deliberately. Their argument — restated in their 1986 paper's caption to Figure 2 — is that antimatter-triggered thermonuclear weapons will have "very

reduced radioactive fallout and electromagnetic pulse effects” because they dispense with the fission primary that produces most of the fallout of a conventional thermonuclear weapon. They further note that this reduced-fallout property is, from a non-proliferation standpoint, itself a problem: “the possible advent of extremely compact and essentially clean nuclear weapons would further diffuse the distinction between low-yield nuclear weapons and conventional explosives.” This is the framing the Bulletin, IEER, and FAS subsequently adopted. We use the “environment-friendly” terminology in this report in that analytical spirit — not as a positive endorsement of the underlying technology.

Chapter 2 Physics of Antimatter-Induced Thermonuclear Ignition

This chapter reproduces the core physics arguments of Gsponer & Hurni (1986) and subsequent refinements (Perkins-Orth-Tabak 2004; Shmatov 2005; Azizi 2023). All equations and numerical values are taken directly from the cited open literature; no original derivations are claimed.

2.1 Antiproton annihilation channels

Proton-antiproton annihilation proceeds through many final-state channels. The dominant average channel — first identified by Fermi and Teller (1947) and confirmed by extensive CERN and LEAR measurements — produces three charged pions and four photons (the photons arising from prompt π^0 decay):

$$p + \bar{p} \rightarrow 3\,\pi^{\pm} + 4\,\gamma$$

with average energies $E_{\pi^{\pm}} = 236$ MeV and $E_{\gamma} = 187$ MeV per annihilation, for a total of 1876 MeV released per annihilation event. Antiproton annihilation on a neutron proceeds similarly. The three charged pions are the workhorse for plasma heating because they are charged and therefore deposit their kinetic energy in the surrounding fuel by multiple Coulomb interactions with electrons. The 187 MeV photons, by contrast, have a 25 m mean free path in solid-density hydrogen and largely escape the fuel — they do not contribute significantly to heating.

The charged-pion energy-loss rate in solid hydrogen-density DT is approximately dE/dx = 0.52 MeV/cm; in Li_2DT (lithium-deuteride-tritide) it rises to 2.06 MeV/cm because of the higher electron density. This difference is one of two reasons Li_2DT is preferred as the thermonuclear fuel for antimatter-catalyzed ignition: it extracts roughly four times more pion energy per unit path length than DT. The second reason is that lithium breeds tritium in situ under neutron bombardment via the $^6Li(n,\alpha)T$ reaction, sustaining the burn.

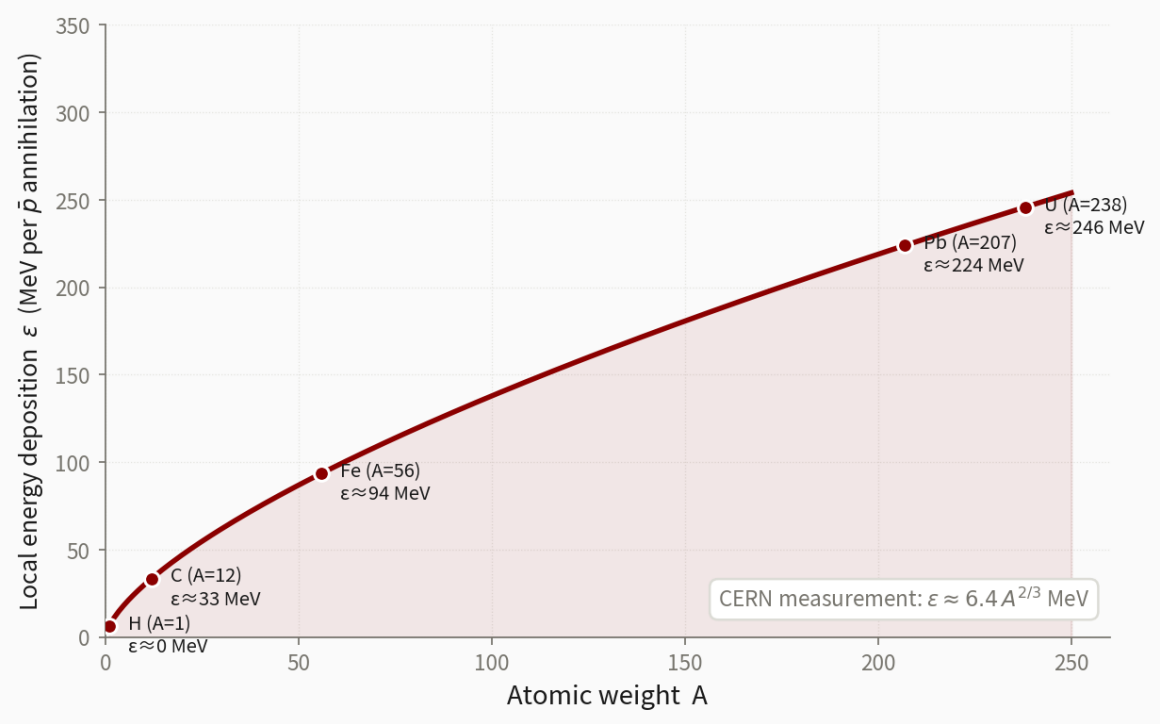


Figure 2.1 — Local energy deposition ε per antiproton annihilation on a nucleus of atomic weight A. CERN measurement (Sullivan 1985) confirms ε ≈ 6.4 $A^{2/3}$ MeV. Heavy-nucleus doping (U-238, A=238) yields ~33× more local deposition than DT alone, which is the basis of Shmatov's reduced antiproton-count estimates.

2.2 Compression factor and the $N \approx 10^{21}/\kappa^2$ scaling

The spark model of thermonuclear ignition requires two conditions to be met simultaneously in a small central region (the "spark"): (a) a critical amount of energy E_c must be deposited in the spark volume, and (b) the temperature in that volume must exceed a critical value T_c. For uncompressed solid DT, Gsponer & Hurni give $E_c = 5\times10^{25}$ keV and $T_c = 4$ keV; for Li_2DT, $E_c = 3\times10^{26}$ keV and $T_c = 13.6$ keV. When the fuel is compressed by a factor κ, both critical quantities scale down as $1/\kappa^2$ because the spark volume shrinks and the density (and thus the Coulomb-energy-deposition rate per unit path) rises.

The antiproton-count requirement follows from coupling the spark energy condition to the annihilation energy budget. Equation (9) of the source PDF gives:

$$E_c / \kappa^2 = N \cdot (\nu \, dE/dx \, \kappa \, R_s + \varepsilon)$$

where ν = 3 (charged pions per annihilation), dE/dx is the pion stopping power in uncompressed fuel, κ is the compression factor, R_s is the spark radius, and ε is the local energy deposition from the recoiling nucleus and pion-nucleus cascade debris. The temperature condition, Equation (10) of the source, is:

$$E_c / \kappa^2 = (3/2) \, z \, a \, \kappa \, \rho \, N_A \, (4\pi/3) \, R_s^{\,3} \, kT_c$$

with z, a = 2, 2.5 for DT and 6, 9.5 for Li_2DT. Solving the two equations simultaneously for N and R_s at κ = 30 (a modest compression factor achievable with chemical explosives), Gsponer & Hurni obtain $N = 3\times10^{18}$, R_s = 0.09 cm for DT and $N = 6\times10^{18}$, R_s = 0.07 cm for Li_2DT. They adopt 10^{18} antiprotons as a working figure sufficient to trigger the burn wave in either fuel at κ = 30.

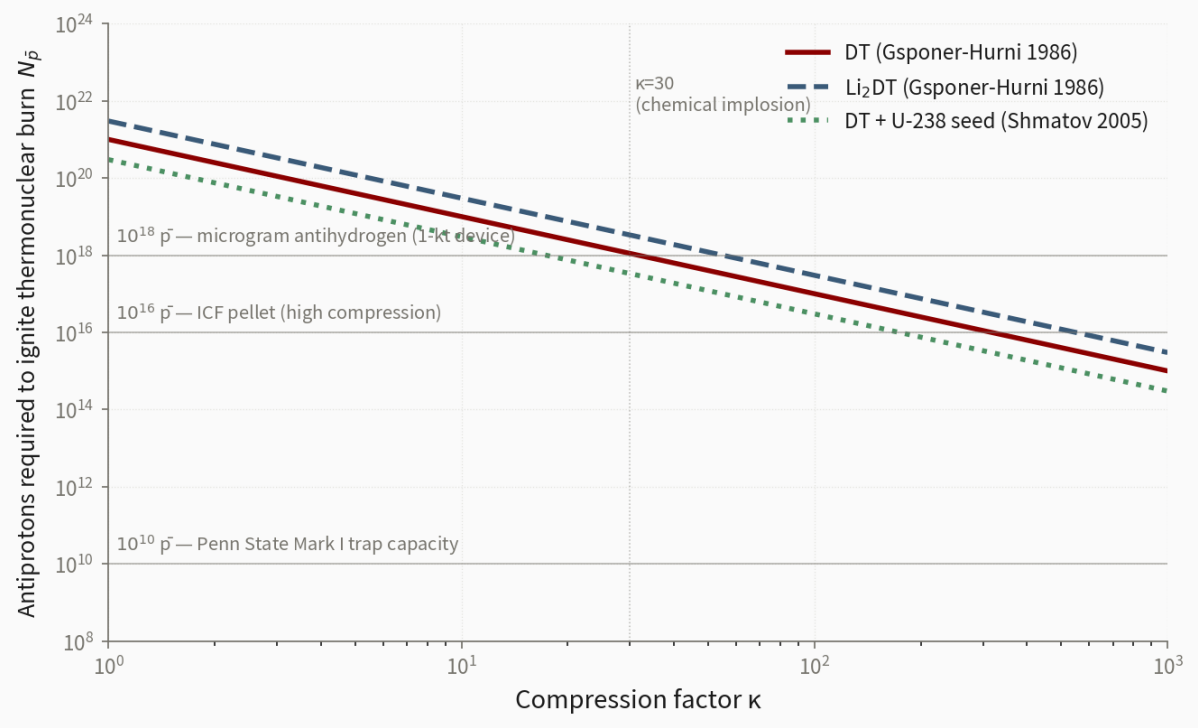


Figure 2.2 — Antiproton count required to ignite a thermonuclear burn wave, as a function of compression factor κ. Three literature estimates are shown: Gsponer-Hurni for DT and Li_2DT (1986) and the Shmatov refinement for DT doped with U-238 (2005). The horizontal reference lines mark the 10^{18} antiproton threshold for a 1-kt antimatter device, the 10^{16} threshold relevant to high-compression ICF, and the 10^{10} storage capacity of the Penn State Mark I trap. Production rate is the binding constraint, not physics.

2.3 The U-238 doping refinement (Shmatov 2005, Azizi 2023)

The CERN measurement of antiproton annihilation in light nuclei (Sullivan 1985) gave a local-deposition term $\varepsilon \approx 6.4\ A^{2/3}$ MeV — a result compatible with the surface-annihilation picture in which low-energy antiprotons annihilate primarily on the nuclear surface rather than penetrating to the core. For DT fuel ($\varepsilon \approx 12$ MeV per annihilation, half the Fermi energy), this term is a small contribution to the spark energy. For heavy nuclei, however, ε rises sharply: ~33 MeV in carbon, ~100 MeV (theoretical, LANL) to ~33 MeV (measured, CERN) in lead, and substantially more in uranium. Shmatov's 2005 JBIS paper argued that doping the DT hot-spot with U-238 would therefore reduce the antiproton requirement by a factor of order 10 over the Gsponer & Hurni estimate, because the antiproton-uranium annihilation would deposit tens of MeV locally — much more efficiently than pion-mediated heating alone.

Azizi et al. (Physica Scripta 98, 2023) extended this analysis to a fast-ignition ICF configuration in which an antiproton beam drives ignition in a DT target seeded with a U-238 layer, and reported a significant gain improvement over pure DT fast ignition. The 2005 and 2023 results together establish that the Gsponer & Hurni antiproton-count estimate is conservative for engineering purposes — actual requirements are likely to be lower once heavy-nucleus seeding is included. The implication for non-proliferation is important: the antiproton-count requirement, while still far beyond current production, is not a fixed physical limit but an engineering-optimization variable.

2.4 Burn-wave propagation

Once the spark conditions (a) and (b) are met, an outgoing spherical thermonuclear detonation wave self-propagates into the surrounding fuel. The temperature profile evolves as a Gaussian-like front centred on the burn-wave radius that advances outward at a velocity set by the α-particle re-deposition length and the fuel's areal density ρR. For Li_2DT at κ = 30, the burn wave consumes the fuel in a few nanoseconds, releasing the full thermonuclear yield in a microscopic fraction of the chemical-implosion time scale. This is the same physics that NIF demonstrated on 5 December 2022, except that NIF's spark was created by laser-driven X-ray ablation of a hohlraum rather than by antiproton annihilation.

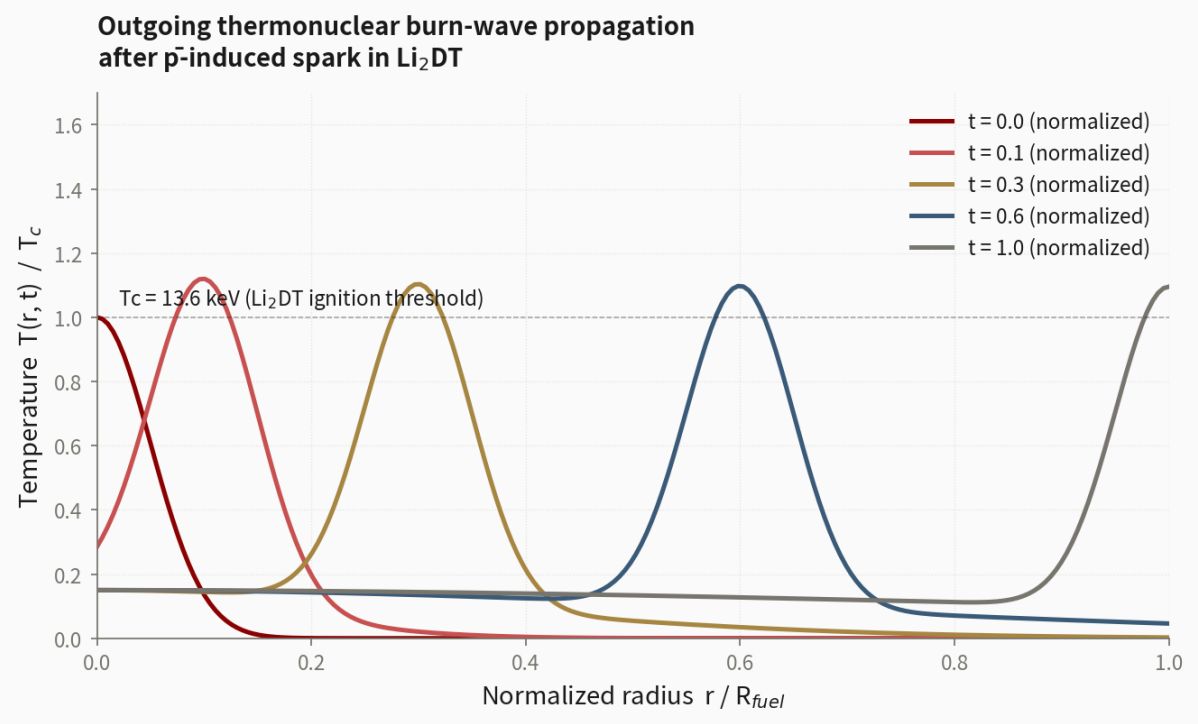


Figure 2.3 — Radial temperature profile in compressed Li_2DT at five successive normalized times after antiproton-induced spark ignition. The Gaussian burn-wave front advances outward; the fuel behind the front is heated to the ignition temperature T_c = 13.6 keV and burns to completion. The temperature scale is normalized to T_c; the radius to the outer fuel radius R_{fuel}. Illustrative model, not a hydrodynamic simulation.

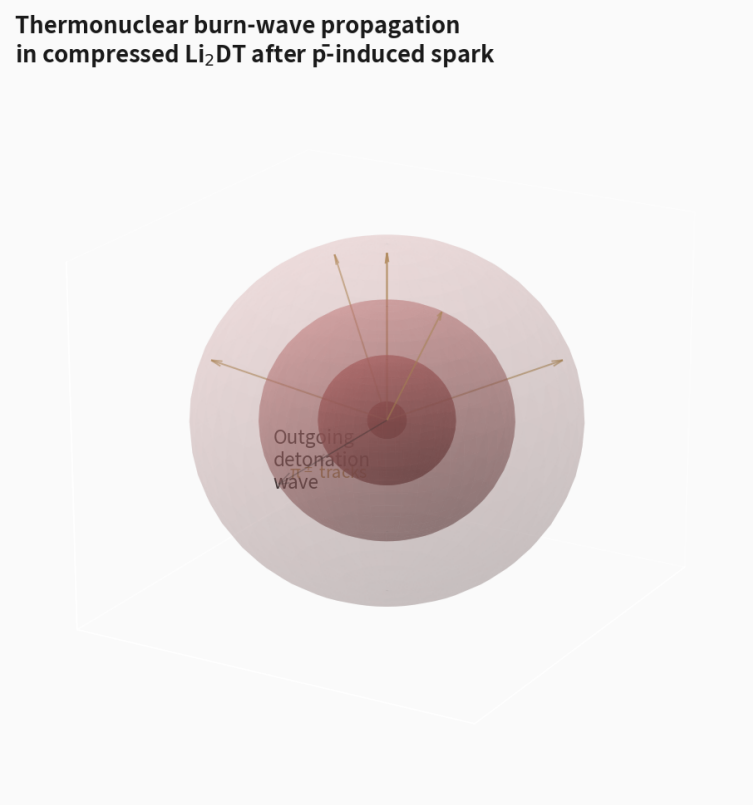


Figure 2.4 — Three-dimensional rendering of the burn-wave propagation. Four nested surfaces represent the spark (red, central), the burn front, the propagating burn wave, and the unburned outer fuel. Yellow spokes emanating from the centre represent the charged-pion tracks that deliver the spark energy from antiproton annihilations to the surrounding fuel.

Chapter 3 State of the Art 2022-2026

This chapter surveys the four enabling-technology streams that have advanced materially since 2022: inertial-confinement fusion at NIF, magnetized target fusion at General Fusion, antimatter handling at CERN, and antiproton-driven fast-ignition theory. The summary table at the end of the chapter collates the antiproton-count estimates across the four open-literature studies.

3.1 Inertial-confinement fusion at NIF

The National Ignition Facility at Lawrence Livermore National Laboratory is the world's largest laser ICF driver: 192 ultraviolet beams delivering up to 2.05 MJ of laser energy to a deuterium-tritium target inside a hohlraum. The facility pursued indirect-drive ICF for over a decade without reaching the ignition threshold. On 8 August 2021, a shot produced 1.32 MJ of fusion yield — the first time the facility came within reach of ignition. On 5 December 2022, the historic ignition shot delivered 2.05 MJ of laser energy and produced 3.15 MJ of fusion energy — a 1.5× target gain, the first controlled-fusion ignition in a laboratory.

NIF has since repeated ignition on more than eleven shots. The February 12, 2024 shot was reported to have achieved 5.2 MJ yield at 2.4× target gain, and an April 7, 2025 shot reportedly reached 8.6 MJ. While these are still microscopic yields on a weapons scale — 8.6 MJ is roughly two kilograms of TNT equivalent, a millionth of a kiloton — they are nuclear explosions in the technical sense: a self-sustaining thermonuclear burn in DT fuel, ignited without fissile material, in a laboratory device.

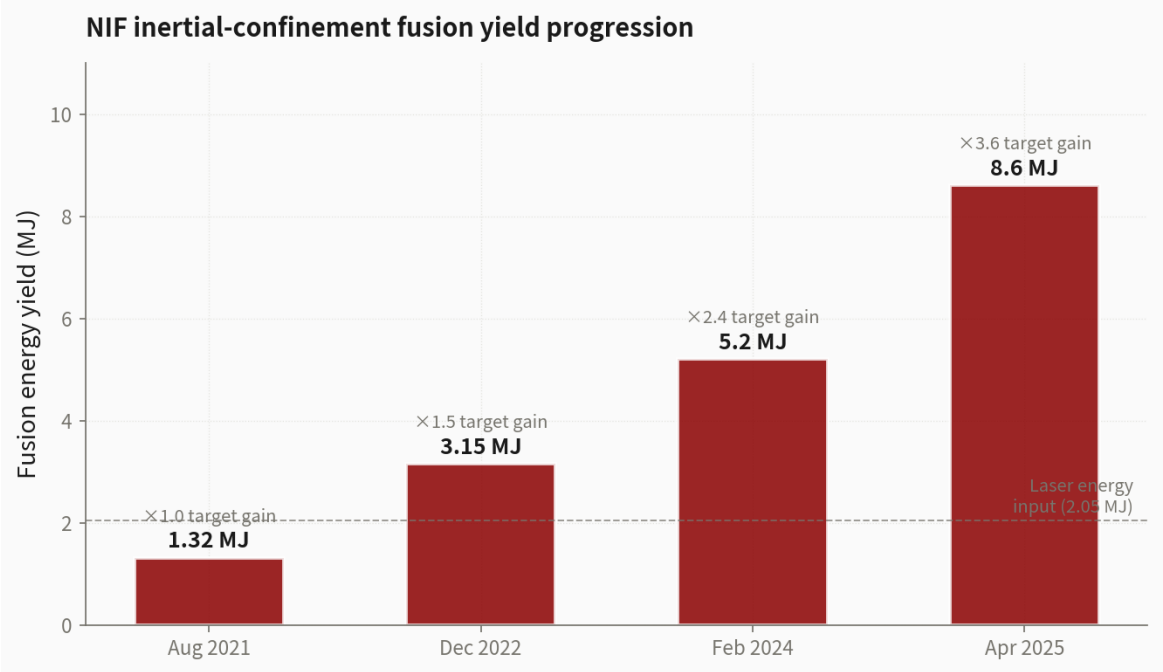


Figure 3.1 — NIF inertial-confinement fusion yield progression, 2021-2025. The Aug 2021 shot reached the ignition threshold; Dec 2022 achieved net target gain; Feb 2024 and Apr 2025 set progressive records. The dashed horizontal line marks the laser-energy input; shots above this line achieve target gain > 1.

The significance of these shots for the antimatter-triggered-fusion agenda is twofold. First, the empirical confirmation that κ = 30-class compression of a DT target produces ignition proves that the spark conditions

Gsponer & Hurni computed in 1986 are physically attainable. Second, the fact that ignition has been reproduced on more than eleven shots — not a one-off anomaly — means the ICF community has crossed from proof-of-principle to operational practice. The remaining obstacle to a pure-fusion device is not the physics of ignition but the engineering of a compact driver.

3.2 Magnetized Target Fusion (General Fusion LM26)

Magnetized Target Fusion (MTF) is the ignition concept first proposed by Sakharov in 1951 and developed at Los Alamos and the Soviet Arzamas-16 lab in the 1990s. The fuel is a magnetized DT plasma (a Field-Reversed Configuration or spheromak) that is preheated to tens of eV and then compressed by an imploding liquid-metal liner driven by pneumatic pistons (the General Fusion approach) or by chemical explosives (the classical weapons-relevant approach). The embedded magnetic field suppresses electron thermal conduction, allowing the fuel to reach ignition conditions with compression ratios much lower than ICF — of order $\kappa = 10$ to 30.

General Fusion's LM26 (Lawson Machine 26), inaugurated in March 2025, is the most advanced MTF device in operation. By late 2024 the machine had achieved stable magnetized plasmas with energy-confinement times exceeding 10 ms, plasma temperatures near 0.72 keV, and sustained fusion-neutron yields of approximately 6×10^{8} neutrons/s (peer-reviewed in *Nuclear Fusion*, 2024 and 2025). LM26's stated roadmap is to reach 1 keV (10 million °C) and then 10 keV (100 million °C) plasma temperatures in successive campaigns, with UKAEA co-authoring the first "world-first" MTF technical milestone in October 2026.

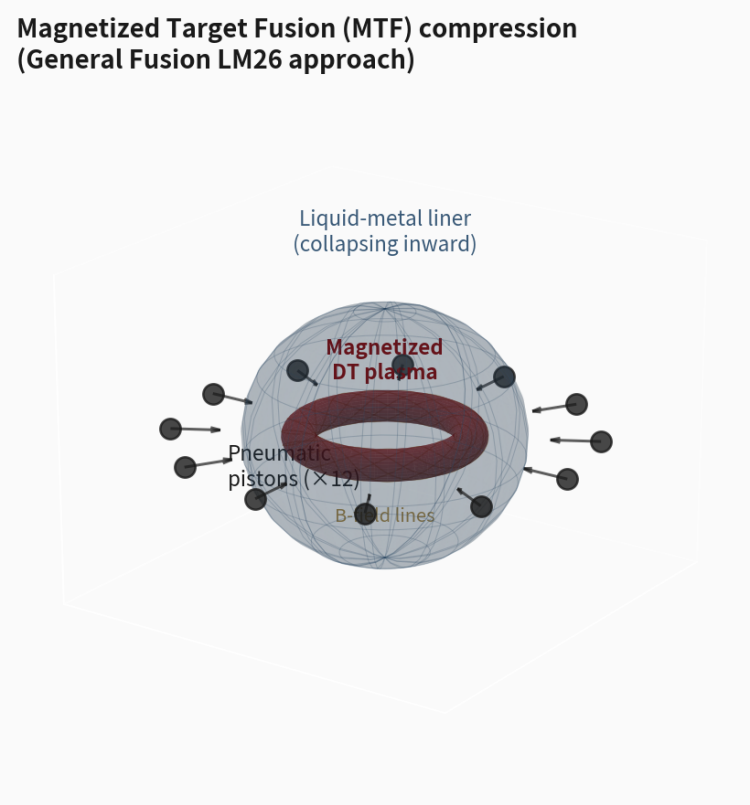


Figure 3.2 — Magnetized Target Fusion compression geometry (General Fusion LM26 approach). A magnetized DT plasma torus is at the centre; the surrounding liquid-metal liner implodes inward under the action of twelve pneumatic pistons arranged around the equator. Helical lines indicate the magnetic field that suppresses thermal conduction during compression.

The arms-control significance of MTF is that it is the only fissile-free ignition route that scales with chemical-explosive compression. The IEER's 1998 report flagged this explicitly: a 1-tonne-class MTF device could in principle be assembled with present-day technology, would yield roughly 1 tonne TNT equivalent, and would kill primarily by neutron radiation with a lethal radius 30 to 800 times that of a chemical explosive of comparable blast yield (Bulletin of the Atomic Scientists, November 2024). The compactness of MTF is the property that makes it the closest-to-weaponizable of the four ignition routes reviewed in this report.

3.3 Antimatter handling at CERN (ALPHA, BASE, BASE-STEP)

Three CERN collaborations have made transformative progress on antimatter handling in 2024-2026. The ALPHA collaboration (antihydrogen spectroscopy and gravity) announced on 18 November 2025 a new sympathetic-cooling technique: a cloud of laser-cooled beryllium ions is added to the positron plasma in the ALPHA trap, cooling the positrons to approximately -266 °C, which makes antihydrogen formation much more efficient when antiprotons are subsequently introduced. The result is an 8× increase in antihydrogen production rate — ALPHA can now accumulate 15,000 antihydrogen atoms in seven hours, versus the previous ten-week cadence. The 2023-24 experimental runs produced over 2 million antihydrogen atoms.

The BASE collaboration (Baryon Antibaryon Symmetry Experiment) reported in 2024 the longest antiproton storage time ever achieved: 600 days in a Penning trap. This compares to the seconds-scale storage demonstrated in 1986 when the Gsponer & Hurni paper was written. In 2025 BASE demonstrated the first coherent quantum-state manipulation of a single antiproton spin (50-second coherence time, the first "antimatter qubit"). The BASE-STEP transportable trap was commissioned in 2024 and moved an unbonded proton cloud across the CERN site; in March 2026 the same transport was accomplished with 92 trapped antiprotons (Leonhardt et al., *Nature*, 2026). The Caravita et al. (arXiv:2503.22471, 2025) AD/ELENA roadmap explicitly lists "transportable antimatter traps" as a goal reaching beyond 2040. Antimatter has, in 2026, become a road-transportable commodity.

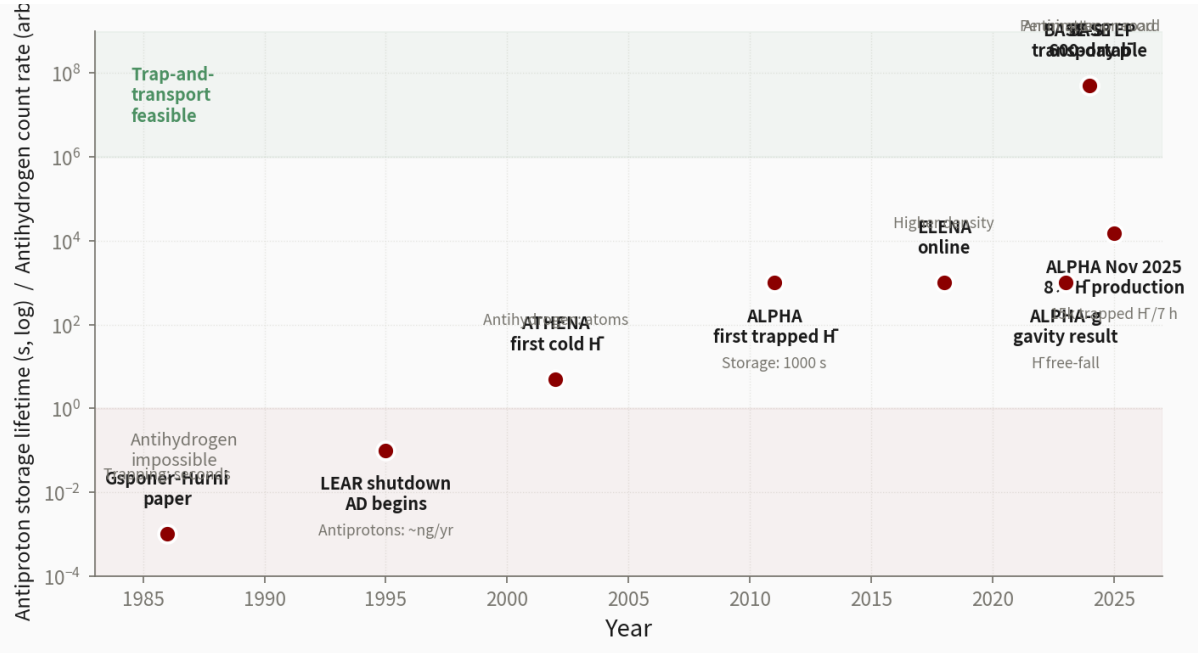


Figure 3.3 — Antimatter handling progress timeline, 1986 to 2025. Antiproton storage lifetime has improved from seconds (1986) to ~600 days (2024 BASE). Antihydrogen production has progressed from first trapped atoms (2011 ALPHA) to 8× faster

production via sympathetic cooling (Nov 2025 ALPHA). The lower shaded band marks the regime where antihydrogen production is impossible; the upper band marks the regime where trap-and-transport is feasible.

3.4 Antiproton-driven fast ignition: 2004 to 2023

The Perkins-Orth-Tabak 2004 paper (*Nuclear Fusion* 44, 1097-1117) is the LLNL follow-up that Gsponer & Hurni cite as confirmation of their original analysis. It introduced Antimatter-Initiated Microfusion (AIM) as a propulsion concept: a cloud of ~10^{10} trapped antiprotons is brought into contact with a D-^{3}He or DT fuel pellet; antiproton annihilation provides the spark energy. The paper emphasizes the unique specific energy of antiprotons — 90 MJ per microgram — which is unmatched by any conventional ICF driver and which makes AIM attractive for deep-space missions where recirculating power is at a premium.

Shmatov (Ioffe Institute, *JBIS* 58, 2005) extended the analysis to a DT hot-spot doped with U-238. His key finding is that the antiproton-uranium annihilation deposits energy locally through the heavy-nucleus cascade, exploiting the $\varepsilon \approx 6.4\ A^{2/3}$ MeV term that Gsponer & Hurni had identified. Shmatov's estimate of the antiproton count required is approximately one order of magnitude lower than Gsponer & Hurni's for comparable ignition conditions. He continued publishing on this theme (IAEA FEC 2014 paper extending to ^{3}He-D hot-spot heating by alpha particles and protons).

The 2023 paper by Azizi et al. (Physica Scripta 98, 095609) reports a computational study of antiproton fast ignition of a DT target with a U-238 seed in a fast-ignition ICF configuration (separate compression and ignition stages). The paper concludes that the antiproton-beam-driven fast-ignition scheme has the potential to significantly improve ignition performance over pure-DT fast ignition. This is the most recent open-literature confirmation that the antiproton route remains intellectually active twenty years after the LLNL AIM paper.

3.5 Cross-literature comparison of antiproton-count estimates

Source (year)	Configuration	Compression κ	N (antiprotons)	Fuel
Gsponer-Hurni 1986	Macro device, chemical implosion	30	~10^{18}	DT or Li_2DT
Gsponer-Hurni 1986	ICF pellet, high compression	300	~10^{16}	DT or Li_2DT
Perkins-Orth-Tabak 2004 (LLNL)	AIM propulsion, cloud + pellet	~10-100	~10^{10} per pulse	D-^{3}He or DT
Shmatov 2005 (Ioffe)	DT hot-spot doped with U-238	30-100	~10^{17} (lowered by U-238)	DT + U-238 seed
Azizi 2023	Fast-ignition ICF, U-238 seed	Fast-ignition geometry	Significant gain vs. pure DT	DT + U-238 seed

Table 3.1 — Open-literature antiproton-count estimates for thermonuclear ignition. All estimates are taken from the cited papers; the "significantly improved" entry for Azizi 2023 reflects that the paper reports relative gain rather than an absolute antiproton count.

Chapter 4 Engineering the Micro and Macro Thermonuke Explosion

Four engineering routes to fissile-free thermonuclear ignition are publicly discussed in the open literature. This chapter defines each, summarizes the engineering state of the art, identifies the special-material dependencies, and compares them on a yield-versus-mass plot. The chapter concludes with an application matrix summarizing the macro/micro explosion regimes.

4.1 Four ignition routes

Route A — Laser-driven ICF (NIF-class).

A 192-beam neodymium-glass laser delivers ~2 MJ of ultraviolet light to a deuterium-tritium target inside a hohlraum. The hohlraum converts the laser light to X rays that ablate the outside of the fuel capsule, driving an implosion to $\kappa \approx 30$-40 and producing a central hot spot at $T > 5$ keV. NIF demonstrated ignition in December 2022. The facility mass is of order 100,000 tonnes (the laser building alone), the energy recirculation is tens of MJ per shot, and the shot rate is at most a few per day. **This route is not weaponizable** in any direct sense — it cannot be miniaturized below building-scale. Its non-proliferation relevance is that it has demonstrated that the physics works and has trained a generation of physicists in ICF hot-spot dynamics.

Route B — Magnetized Target Fusion (MTF).

A magnetized DT plasma (FRC or spheromak, T ~ 50 eV) is compressed by an imploding liquid-metal liner or by chemical explosives. The embedded magnetic field reduces thermal conduction losses so that the fuel reaches ignition at κ = 10-30, much lower than ICF. General Fusion's LM26 is the leading civilian demonstrator (pneumatic-piston liquid-liner compression). The historical weapons-relevant approach uses chemical explosives to drive the liner. MTF scales linearly with chemical-explosive energy, making it the closest-to-compact fissile-free route.

Route C — Antimatter-catalyzed microfusion (ACMF).

Antiprotons annihilate at the centre of a compressed fuel pellet, depositing ~12 MeV per annihilation locally in DT (or up to ~100 MeV in heavy-nucleus-doped fuel). The antiproton count required is $N \approx 10^{18}/\kappa^2$. With κ = 30 (chemical-implosion compression) the requirement is $\sim 10^{18}$ antiprotons (1.67 micrograms of antihydrogen). The binding constraint is antiproton production: worldwide production is ~nanograms per year, four orders of magnitude below the requirement.

Route D — Fast ignition hybrid.

A petawatt-class short-pulse laser heats a pre-compressed fuel core, separating the compression and ignition stages. Fast ignition has been pursued since the early 2000s at ILE Osaka, LLE Rochester, and elsewhere; recent advances in laser-plasma collimation (Takizawa et al., *Physical Review Research* 7, 023081, 2025) have moved it closer to practicality. Fast ignition is the route to which the Azizi 2023 antiproton-beam variant belongs.

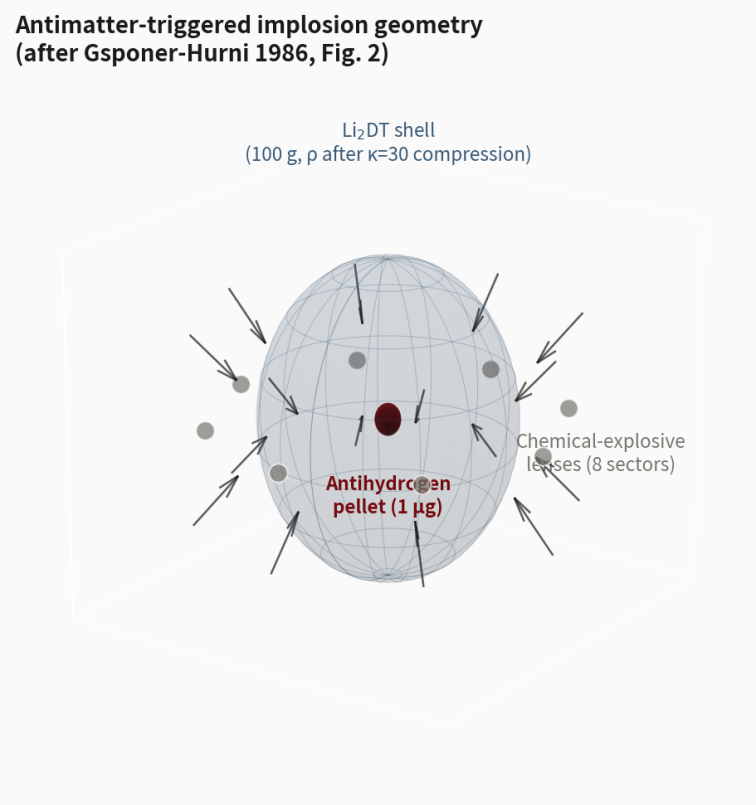


Figure 4.1 — Antimatter-triggered implosion geometry (after Gsponer-Hurni 1986, Figure 2 of the source PDF). A 1-microgram antihydrogen pellet (red sphere, centre) is levitated inside a 100-gram Li_2DT shell (translucent blue). Eight chemical-explosive lenses drive the implosion (gray arrows, inward). At peak compression the shell contacts the antihydrogen, annihilation pions ignite the burn, and a thermonuclear detonation wave propagates outward through the Li_2DT.

4.2 Yield-mass trade-off

Figure 4.2 plots the four ignition routes on a log-log yield-versus-mass diagram, with a modern US thermonuclear warhead included for reference. The key observations are: (a) NIF ICF is a building-scale facility with milliton-equivalent yields — it is many orders of magnitude from a deliverable weapon; (b) MTF sits in the 1-tonne-class, 1-tonne-TNT-equivalent regime that the IEER and Bulletin articles identify as the closest-to-weaponizable pure-fusion concept; (c) the ACMF concept targets a 100-kg-class, 1-kt-class device — if antiproton supply were available, this is the most compact macro-explosion route; (d) fast ignition is an intermediate scale. None of these routes has been demonstrated as a deliverable weapon, and none has a clear path to weaponization on a less-than-decade time scale. The plot makes clear, however, why MTF and ACMF are the routes the arms-control community watches most closely.

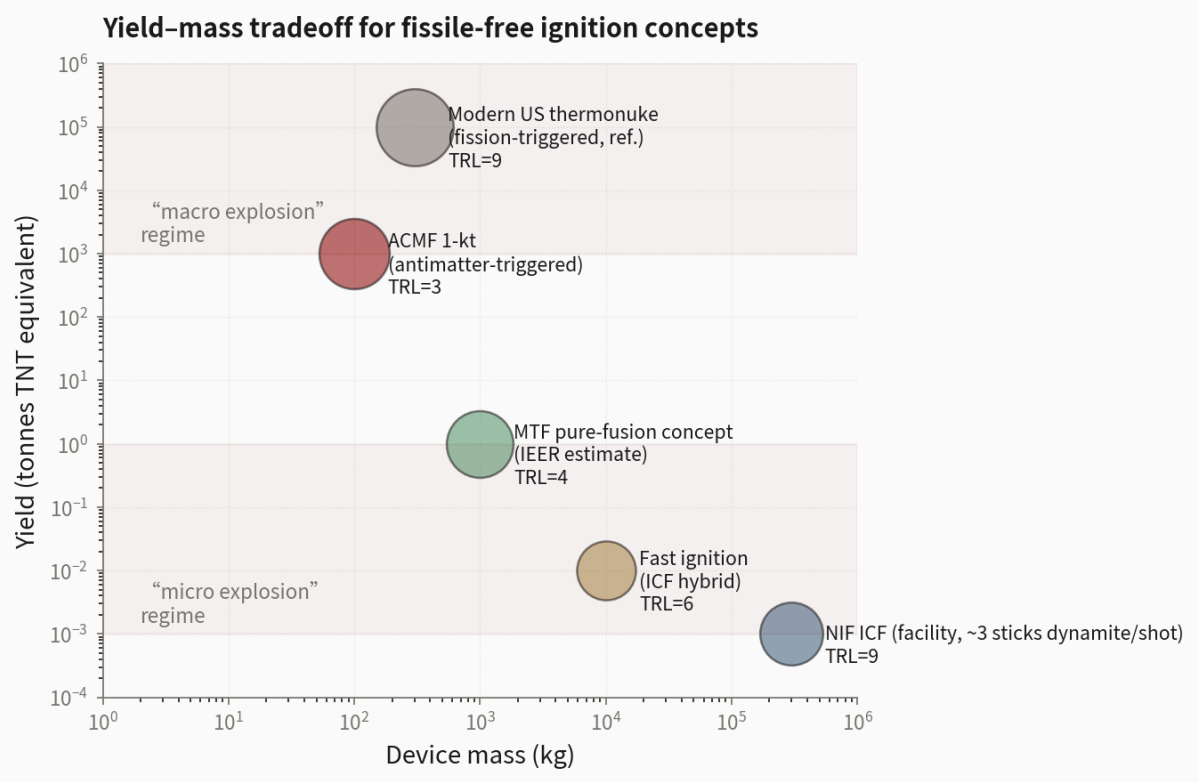


Figure 4.2 — Yield-versus-mass trade-off for the four fissile-free ignition concepts (log-log). Bubble size is qualitative; the modern US thermonuclear warhead is shown for reference. The shaded bands mark the "micro explosion" (below 1 ton TNT equivalent) and "macro explosion" (above 1 kiloton) regimes. MTF sits in the most concerning compact-yield corner.

4.3 Micro versus macro explosions

The distinction between "micro" and "macro" thermonuclear explosions is not merely a yield classification — it has distinct engineering and arms-control implications. Micro explosions (milliton to ton TNT-equivalent) are in the yield range NIF has now demonstrated repeatedly. They are driven by building-scale facilities or, in principle, by MTF devices of a few hundred kilograms. They produce small blast areas but high neutron fluences, making them potentially attractive for radiation-effect weapons or for laboratory study of thermonuclear physics. Macro explosions (kiloton to megaton TNT-equivalent) require the antiproton-catalyzed or MTF-chemically-driven approach to reach weapons-relevant yield, and would constitute, if ever built, a true pure-fusion weapon in the arms-control sense.

Application	Yield range (TNT-eq.)	Driver	Special material	TRL
NIF ICF (laboratory ignition)	~1 kg per shot	Laser (2 MJ UV)	DT, hohlraum	9 (demonstrated)
MTF pure-fusion concept (IEER/Bulletin)	~1 tonne	Chemical explosive or pneumatic liner	DT, lithium, liquid metal	4 (concept validated)
ACMF 1-kt antimatter device	~1 kt	Chemical implosion + antiprotons	Li_2DT, ~10^{18} $\bar{p}$	3 (component-level)

Fast ignition (petawatt-laser)	~1-10 kg per shot	Petawatt laser + compression	DT, cone target	6 (subsystem)
Plasma-jet DEW (Route A, Ch. 5)	~0.1-1 kt (micro)	Shaped MTF or ACMF charge	As above + shaping lens	2 (conceptual)
GRASER pumping (Route B, Ch. 5)	~0.1-1 kt (micro)	ACMF + isomer rod	Nuclear isomer (e.g. ^{178m2}Hf)	2 (conceptual)
AIM propulsion (Route C, Ch. 5)	~0.1-1 kt per pulse (micro)	Magnetic nozzle + antiproton cloud	~10^{10} $\bar{p}$/pulse, D-^{3}He	3 (component)
X-ray laser pumping (Route D, Ch. 5)	~1-10 kt (micro to small macro)	NIF-class laser or ACMF	Lasing medium (e.g. Se, Y)	4 (Teller-Wood legacy)

Table 4.1 — Application matrix. Yield figures are typical open-literature estimates for the indicated configuration; actual yields depend on fuel mass, compression factor, and driver energy. TRL values are this report's assessment based on the cited literature, not official DoD TRL ratings.

Chapter 5 Enabling Directed-Energy Weapons and GRASERs

Gsponer & Hurni's 1986 paper lists "directed energy weapons applications" as one of the prospective uses of antimatter-triggered thermonuclear microexplosions. This chapter elaborates four application tracks: (A) plasma-jet directed-energy weapons driven by shaped thermonuclear microexplosions; (B) GRASERs (gamma-ray lasers) pumped by nuclear-explosion X-ray baths; (C) AIM ultrafast space propulsion; (D) X-ray laser pumping in the Teller-Wood lineage. All four are conceptual — no working device exists. The arms-control dimension is addressed in Chapter 7.

5.1 Track A — Plasma-jet directed-energy weapons (DEWs)

A shaped thermonuclear microexplosion produces, by hydrodynamic focusing, a collimated hypervelocity plasma jet. The principle is identical to a conventional shaped charge, except the energy source is a fusion burn rather than a chemical explosion. The jet velocity is in the 100-1000 km/s range (substantially above the ~10 km/s of chemical shaped-charge jets), the jet temperature is in the keV range, and the jet mass is a fraction of the fusion-fuel mass. The jet deposits its kinetic and thermal energy on a target at distances of order 100-1000 km in the vacuum of space — the range is set by the jet's geometric divergence and its interaction with the residual solar-wind plasma.

Gsponer & Hurni flagged this application in their 1986 paper. The idea is revisited in the broader "directed energy weapons applications may include the trigger of thermonuclear plasma jets" sentence of their Section 7. Open-literature studies of the underlying physics are limited; the concept appears most developed in the Russian literature on cumulative jets from thermonuclear microexplosions and in the U.S. SDI-era literature on nuclear-explosion-pumped lasers and directed plasma. No public-source engineering study gives a working design, and the concept's TRL remains at 2 (concept formulation).

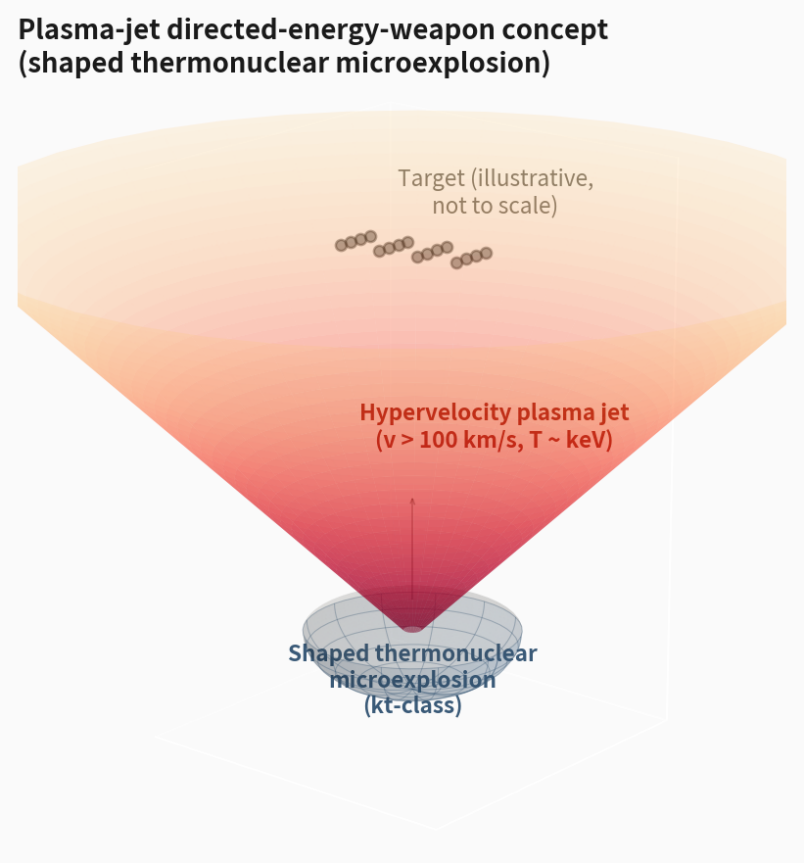


Figure 5.1 — Plasma-jet directed-energy-weapon concept. A hemispherical thermonuclear microexplosion (lower, kt-class, blue wireframe) drives a shaped-charge-like jet of hypervelocity plasma (yellow-orange cone, v > 100 km/s, T ~ keV) toward a target (upper, illustrative, not to scale). The jet's collimation is set by the charge geometry; the jet's range in vacuum is set by hydrodynamic divergence and interaction with the solar-wind plasma.

5.2 Track B — GRASERs (gamma-ray lasers)

A GRASER (Gamma-Ray Amplification by Stimulated Emission of Radiation) is a hypothetical device in which a population inversion in a nuclear transition — typically a nuclear isomer — is pumped by an external energy source and amplifies a seed gamma-ray beam coherently. The 1990s and 2000s saw substantial interest in the hafnium isomer ^{178m2}Hf as a candidate lasing medium because of its 2.45 MeV transition with a 31-year half-life. Claims of triggered release from ^{178m2}Hf by Collins et al. in 1999 could not be reproduced and are not currently accepted; the 2003 DARPA-sponsored review concluded that triggered isomer release was not demonstrated.

The thermonuclear-explosion-pumped variant — the conceptualization that would matter for this report — uses the X-ray bath from a fusion microexplosion to populate an upper nuclear state of an isomer rod, with end mirrors forming a resonator. The energy density required to drive a nuclear transition to inversion is orders of magnitude beyond chemical sources, but is within reach of a thermonuclear microexplosion. Edward Teller's interest in nuclear-pumped gamma-ray lasers is documented in his later-career publications. No working GRASER has ever been built; the concept remains at TRL 2. The companion HTML file accompanying this PDF includes an interactive Three.js visualization of the GRASER pumping-cavity concept.

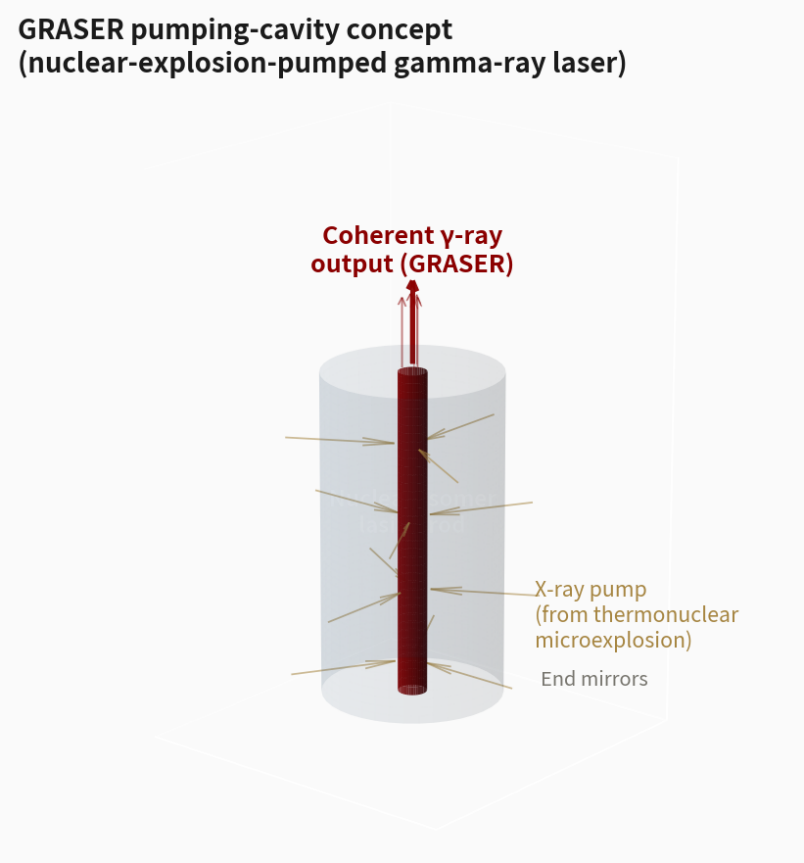


Figure 5.2 — GRASER pumping-cavity concept. A nuclear-isomer lasing rod (red, central) sits in a cylindrical cavity with end mirrors. A thermonuclear microexplosion outside the cavity floods the cavity with X rays (tan arrows, inward) that pump the isomer to its upper state; stimulated emission produces a coherent gamma-ray beam that exits through the front mirror (red arrow, outward). No working device has ever been built.

5.3 Track C — AIM ultrafast space propulsion

The Antimatter-Initiated Microfusion (AIM) concept, developed at Pennsylvania State University in the 1990s and studied at Lawrence Livermore since 2004, is the propulsion application of the antimatter-catalyzed fusion physics. The Penn State Mark I trap can store 10^{10} antiprotons for ~168 hours. In the AIM propulsion variant, a cloud of ~10^{10} antiprotons is brought into contact with a D-^{3}He or DT pellet in a magnetic-nozzle reaction chamber; the resulting microexplosion (yield ~0.1-1 kt TNT equivalent) is directed through the magnetic nozzle, producing exhaust velocities of order 10^{4} km/s and specific impulses of 10^{5}-10^{6} s — two to three orders of magnitude above chemical rockets.

The RAND 1985 study (Augenstein, N-2302-AF/RC) gives a cost estimate of $500 million to $1 billion for a prototype antiproton factory producing 10 to 100 micrograms per year, and $5 to $15 billion for a full production factory of 10 mg per year. The Project Icarus estimate (Wikipedia, 2024) of $100 billion per milligram reflects the gap between the RAND numbers and present-day cost experience. Either way, the cost-per-microgram of antiproton production is far beyond what is needed for an economically viable propulsion system; AIM propulsion is currently a research concept, not a near-term mission architecture.

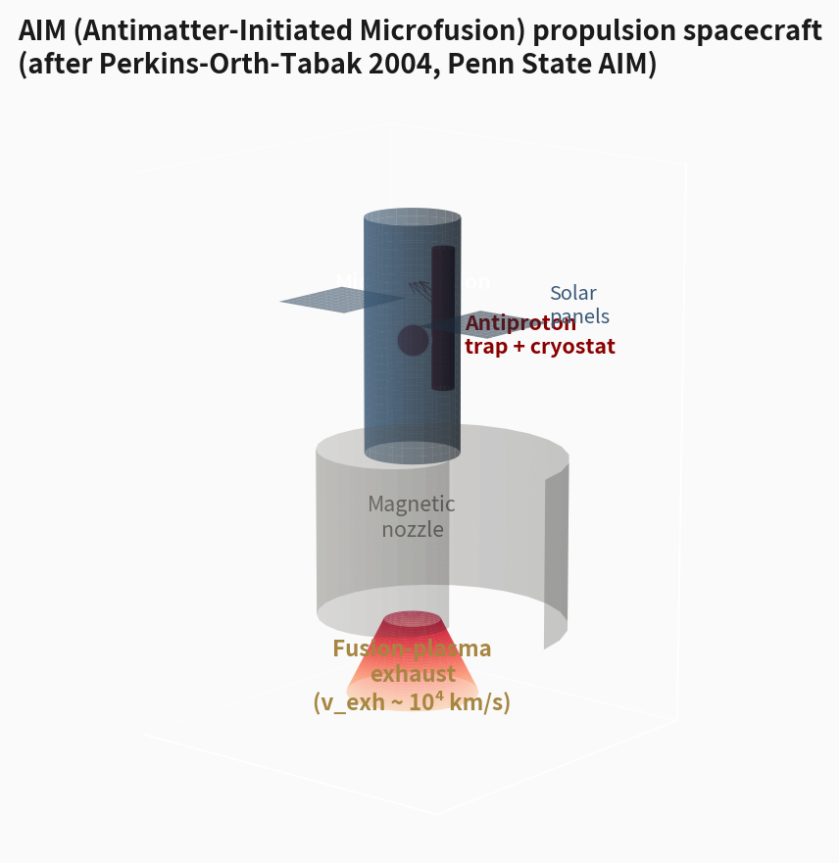


Figure 5.3 — AIM (Antimatter-Initiated Microfusion) propulsion spacecraft. The main body (blue cylinder) contains the microexplosion chamber and the antiproton storage trap (red cylinder on top). A magnetic nozzle (gray cone, rear) directs the fusion-plasma exhaust (yellow-orange gradient). The exhaust velocity is of order 10^4 km/s, giving specific impulse of 10^5-10^6 s. Solar panels provide station-keeping power. After Perkins-Orth-Tabak (2004) and the Penn State AIM concept.

5.4 Track D — X-ray laser pumping (Teller-Wood lineage)

The nuclear-explosion-pumped X-ray laser concept, pursued by Teller and Wood in the 1980s under the SDI program, used a thermonuclear microexplosion to pump population inversions in lasing media (typically selenium or yttrium plasmas) to produce coherent X-ray beams. The 1980s program was declassified in the 1990s after yielding no working device; the basic physics was sound but the engineering of a stable cavity was not achievable in the available time. The NIF-class laser facilities of the 2010s and 2020s have revived interest in laboratory-scale X-ray-laser pumping without nuclear explosions, and the underlying physics of nuclear-pumped X-ray lasers remains of academic interest. This is the route for which the ICF advances of 2022-2025 are most directly enabling.

The four application tracks together span the range from purely conceptual (Track A plasma-jet DEW, Track B GRASER) through component-validated (Track C AIM propulsion) to subsystem-level demonstrated (Track D X-ray laser pumping using NIF-class lasers). None of the four is at TRL 6 or above. The 2022-2025 NIF ignition and the 2024-2026 LM26 MTF milestones have advanced the underlying physics base but have not, in themselves, produced any of the four application devices.

Chapter 6 R&D; Roadmap and TRL Assessment

This chapter lays out a technology-readiness-level (TRL) roadmap for the four enabling subsystems of the application tracks: (1) antiproton production at scale; (2) antihydrogen storage and transport; (3) implosion uniformity for ICF/MTF-class compression; (4) fast-ignition driver; and the four application tracks themselves. The TRL scale used here is the standard NASA / DoD 1-9 scale. We do not assign official TRL ratings — those would require classified or program-internal information not available in open sources — but the qualitative assessments follow directly from the public literature reviewed in Chapters 3 and 5.

6.1 Three enabling-technology streams

(1) Antiproton production at scale.

The current worldwide antiproton production rate is of order nanograms per year, concentrated at CERN (AD/ELENA). The RAND 1985 estimate for a 10-100 μg prototype factory was $500M-$1B; the full 10 mg/year factory was estimated at $5-15B. The Project Icarus estimate for the cost per milligram is $100 billion. No facility approaching the prototype-factory scale exists or is under construction. The antiproton production cross-section measurement program at the AMBER facility at CERN (2023-2024) is the first dedicated measurement of production cross-sections since the 1990s, but it does not constitute a production-scale-up. **This is the binding constraint for all antimatter-triggered application tracks.**

(2) Antihydrogen storage and transport.

ALPHA's November 2025 sympathetic-cooling breakthrough (8× production rate increase) and BASE's 600-day antiproton storage are the two most transformative single advances of the 2024-2026 period. The BASE-STEP transportable trap, with road transport of 92 antiprotons demonstrated (Leonhardt et al., Nature 2026), removes the geographical-binding constraint: antimatter can now leave the production site. The remaining engineering challenges are (a) scaling storage from 10^4-10^5 antiprotons (current) to 10^{10}-10^{12} (Penn State Mark I trap, projected); (b) the materials-science challenge of levitating a frozen antihydrogen pellet in a 1 mm cryostat inside a complex implosion device (Gsponer & Hurni, 1986, Section 7 — "a tremendous challenge for materials microtechnology"); and (c) the long-term maintenance of coherence in transportable Penning traps.

(3) Implosion uniformity and the ICF/MTF driver.

Symmetric thermonuclear ignition requires implosion uniformity at the 1% level across the fuel surface. NIF's 2022 ignition was achieved after a decade of incremental improvements to hohlraum design, target fabrication, and beam balancing. MTF faces a different but related challenge: the liquid-metal liner must remain azimuthally uniform while imploding. General Fusion's LM26 addresses this by using pneumatic pistons that are individually controllable, with feedback from plasma diagnostics. The November 2024 results (6×10^8 neutrons/s sustained) demonstrate that the uniformity is achievable at the plasma-stability level, though not yet at the full ignition-compression level. The Caravita et al. (2025) AD/ELENA roadmap explicitly flags 2040+ timelines for antimatter-trap maturity.

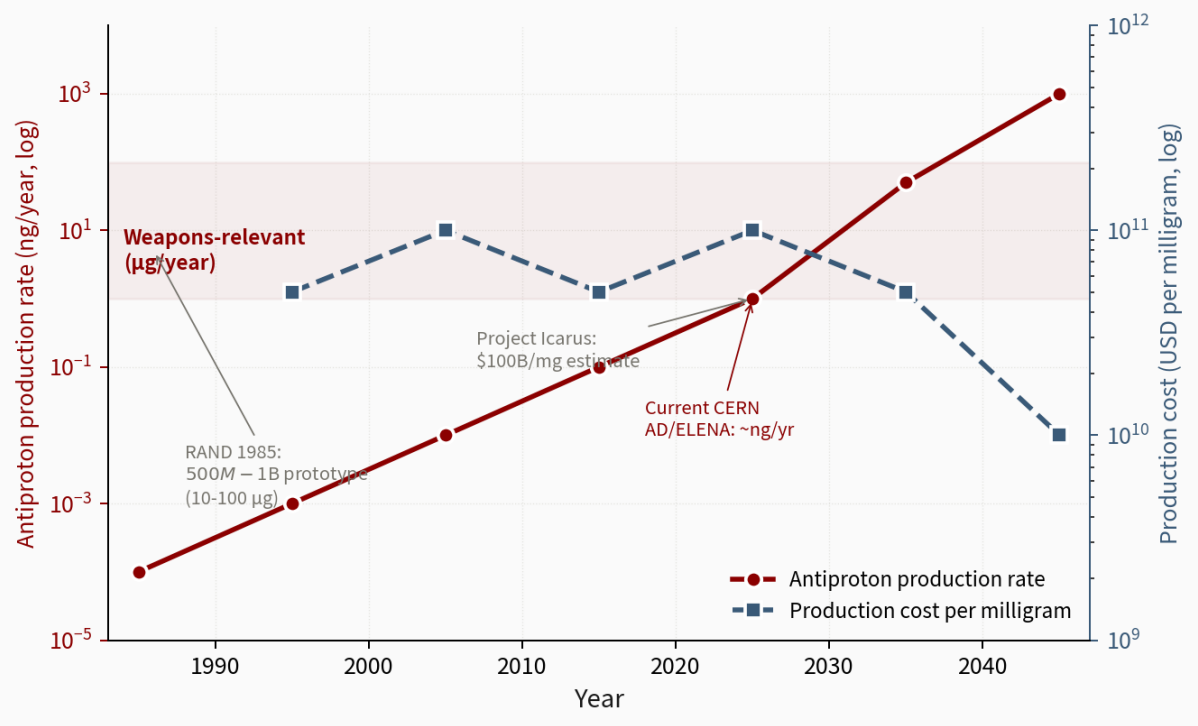


Figure 6.1 — Antiproton production rate and cost projections, 1985-2045. The left axis (red) shows production rate in nanograms per year; the right axis (blue) shows estimated production cost per milligram in USD. The shaded band marks the "weapons-relevant" production-rate regime (micrograms per year and above). Current production is approximately four orders of magnitude below the weapons-relevant threshold; even the 2045 projection does not close the gap.

6.2 TRL roadmap by subsystem and application

Figure 6.2 summarizes the TRL roadmap. The horizontal bars span the period from first demonstration (or current state) to the projected target TRL for each subsystem or application track. The vertical red dashed line marks 2026, the date of this report. The bars are shaded so that the dark portion represents already-achieved TRL progress and the light portion represents the projected path to the target TRL. Several observations follow from the figure: NIF-class ICF ignition is at TRL 9 and stable; fast ignition is at TRL 6 with a path to TRL 8 within a decade; MTF is at TRL 4 with a path to TRL 7 by the mid-2030s; antiproton production scale-up is at TRL 3 with a path to TRL 6 only by the early 2040s; antimatter-catalyzed microfusion and the four application tracks are at TRL 2-3 and will not reach TRL 6 before the late 2030s at the earliest.

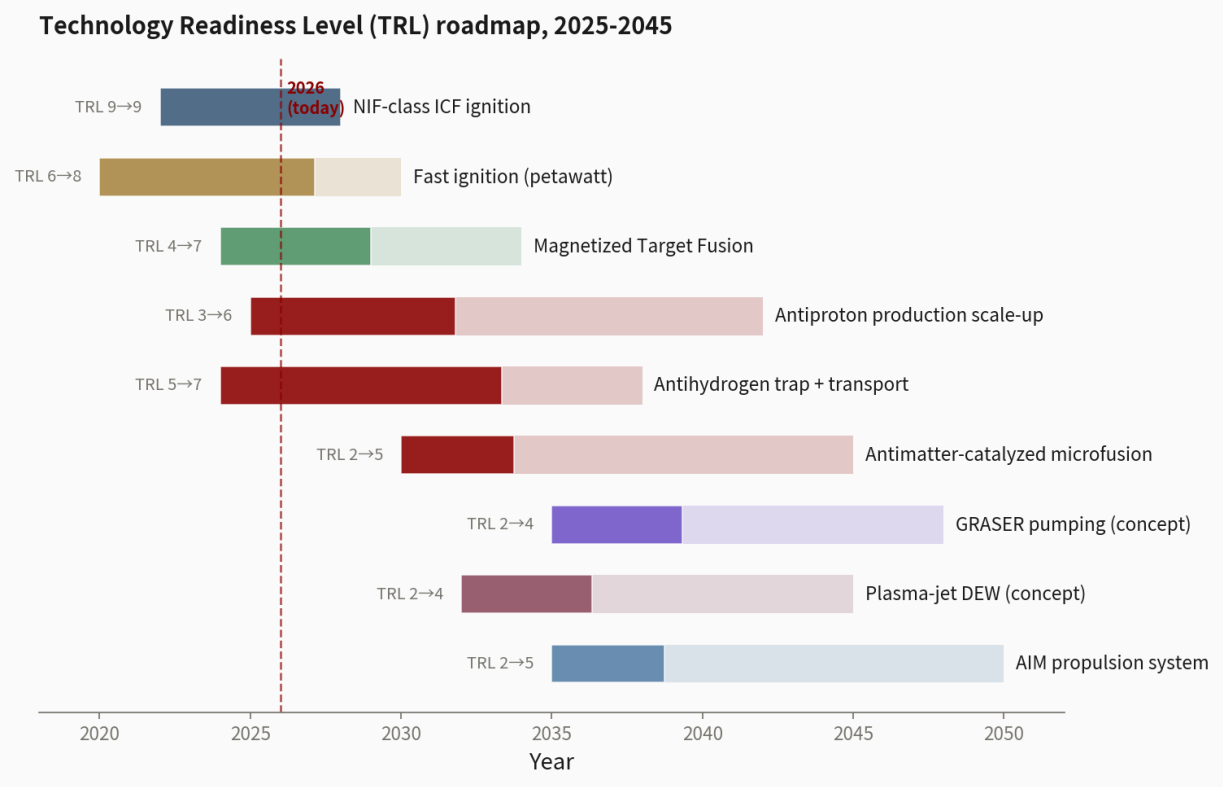


Figure 6.2 — Technology-readiness-level (TRL) roadmap, 2025-2045. Nine fusion-and-antimatter programs are shown as horizontal bars. Dark shading marks already-achieved TRL progress; light shading marks projected progress to the target TRL. The vertical dashed red line marks 2026. NIF-class ICF ignition is the only TRL-9 technology; fast ignition and antihydrogen trap and transport are at TRL 5-6; all four application tracks are at TRL 2-3.

Chapter 7 Arms Control and Non-Proliferation

This chapter summarizes the arms-control implications of the technological developments reviewed in Chapters 3-6. The analysis follows the open-literature framework established by Jones & von Hippel (*Science & Global Security* 6, 1998) and the IEER (*Dangerous Thermonuclear Quest*, 1998), updated with the Bulletin of the Atomic Scientists November 2024 article *The entanglement of fusion energy research and bombs*. No original policy analysis is claimed.

7.1 CTBT Article I scope and the 2022 NIF shot

Article I of the Comprehensive Test Ban Treaty (CTBT) prohibits "any nuclear weapon test explosion or any other nuclear explosion." The "any other nuclear explosion" clause is deliberately broad: it covers peaceful nuclear explosions, subcritical tests of nuclear-fission assemblies, and (in principle) pure-fusion explosions of any yield. The CTBT has not entered into force — it requires ratification by 44 Annex-2 states, of which eight have not ratified (China, Egypt, India, Iran, Israel, North Korea, Pakistan, United States) — but the major nuclear powers observe its prohibitions voluntarily.

The Bulletin of the Atomic Scientists' November 2024 article argues that the December 2022 NIF ignition shot — which released 3.15 MJ of fusion energy in nanoseconds — is technically a "nuclear explosion" under the CTBT Article I scope. The article notes that the energy released was equivalent to roughly three sticks of dynamite; that the energy was released in nanoseconds, the time scale of a thermonuclear burn; and that the device was ignited without fissile material. The argument is contested — the U.S. government's position is that NIF experiments are stockpile-stewardship research, not nuclear-explosive testing — but it raises the first serious public question of whether civilian ICF research is bounded by the CTBT.

7.2 The verification gap created by the absence of fissile material

The Nuclear Non-Proliferation Treaty (NPT) and the IAEA safeguards system that implements it are organized around fissile material: highly-enriched uranium and separated plutonium are the "special nuclear materials" whose production, processing, and use trigger safeguards obligations. A pure-fusion weapon requires no special nuclear material. The deuterium and tritium that fuel it are not subject to IAEA safeguards in the same way: tritium is a "by-product material" and deuterium is not safeguarded at all. The NPT's verification architecture has no direct hook for a pure-fusion device.

This is the verification gap the IEER's 1998 report and the 2024 Bulletin article identify as the most consequential long-term effect of pure-fusion research. Even if no pure-fusion weapon is ever built, the existence of an industrial base that could build one — the 140+ fusion facilities in 50+ countries identified by the Bulletin article — makes the existing safeguards regime increasingly leaky. Antimatter production facilities present an analogous problem: the CERN AD/ELENA is not safeguarded (it produces far too little antimatter to matter), but a scaled-up antiproton factory of the kind RAND estimated in 1985 would have no automatic safeguards hook.

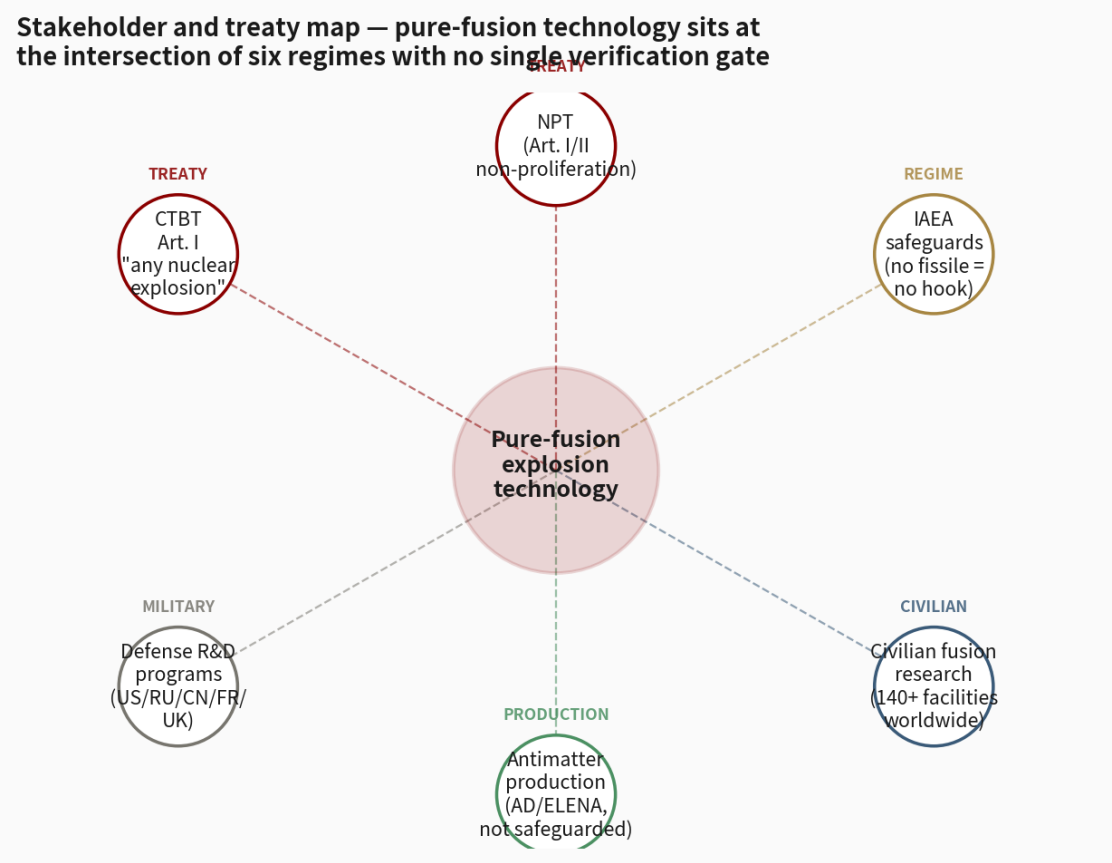


Figure 7.1 — Stakeholder and treaty map. Pure-fusion technology sits at the intersection of six regimes — CTBT, NPT, IAEA safeguards, civilian fusion research, antimatter production, and defense R&D; programs — none of which provides a single verification gate that comprehensively covers it.

7.3 Policy options proposed in the open literature

The open arms-control literature has advanced three families of policy options for the pure-fusion challenge:

(1) No-first-pure-fusion-weapon declarations. The Bulletin article recommends that the nuclear-weapon states make unilateral declarations that they will not develop, test, or field pure-fusion weapons. This would extend the no-first-use logic to a new technology class without requiring treaty renegotiation. The political cost is low; the verification burden is essentially zero (since pure-fusion weapons do not yet exist).

(2) Threshold-yield declarations for ICF/MTF facilities. Jones & von Hippel (1998) and the IEER (1998) recommend that states with major ICF or MTF facilities (NIF, LMJ, LM26, Shenguang, UFL-2M, etc.) declare that they will not conduct fusion-explosion experiments above a stated yield threshold (e.g. 1 ton TNT equivalent). This would create a transparency regime for the ICF research community without banning it, and would provide a verification hook (the yield can be remotely estimated from the prompt-neutron signature).

(3) Safeguarding antimatter-production facilities above a threshold. The Gsponer & Hurni paper (Section 7, 1986) and the IEER report both recommend that any future antiproton-production facility above a stated production-rate threshold (e.g. 1 mg/year) be subject to international safeguards. This would extend the IAEA's "additional protocol" logic to a new facility type. The threshold is important: present-day AD/ELENA production (~ng/year) is far below any weapons-relevant level and need not be safeguarded.

7.4 The closest-term policy concern is MTF, not antimatter

The single most important policy conclusion of this report is that the closest-term pure-fusion threat is magnetized target fusion, not antimatter-catalyzed microfusion. Antimatter production is at least two decades from weapons-relevant rates; MTF is at TRL 4 today with a clear engineering path to TRL 7 within a decade. The LM26 program is openly civilian but the underlying compression physics is identical to a weapons-relevant device. The IEER/Bulletin argument that a 1-tonne-class MTF pure-fusion device is deliverable with present-day technology is not a far-future scenario; it is an engineering assessment based on the published LM26 results. Arms-control policy attention should focus on MTF and on threshold-yield declarations for ICF facilities, rather than on the more exotic but more distant antimatter-facility safeguarding question.

Chapter 8 Conclusions and Outlook

Three material changes have occurred in the open literature on fissile-free thermonuclear ignition since Gsponer & Hurni's 1986 paper. This report has surveyed them and laid out the implications for R&D; planning and arms control. The conclusions can be summarized in three sentences:

(1) The ICF hot-spot has been ignited. NIF's December 2022 shot and the eleven subsequent ignition shots have demonstrated, reproducibly, the thermonuclear burn-wave conditions that Gsponer & Hurni computed in 1986 for an antimatter-catalyzed spark. The spark-energy figure $E_c = 3\times10^{26}$ keV for Li_2DT (and the scaling with $1/\kappa^2$) is no longer a theoretical estimate; it is an empirical boundary condition met by a laboratory device.

(2) Antimatter handling has gone from seconds to 600 days. The BASE 600-day antiproton storage (2024), the ALPHA November 2025 eight-fold antihydrogen production breakthrough, and the BASE-STEP road-transportable trap (Leonhardt et al., Nature 2026) have collectively moved antimatter from a curiosity to an industrial-class handling capability. The 10^{18} antiprotons Gsponer & Hurni estimated as the requirement for a 1-kt device is still twelve orders of magnitude above current worldwide production, but no longer a fundamental physics obstacle; it is a manufacturing-scale obstacle.

(3) Magnetized Target Fusion has emerged as a compact, non-fission, non-antimatter ignition route. General Fusion's LM26 machine, with plasma temperatures near 0.72 keV, confinement times exceeding 10 ms, and sustained neutron yields of 6×10^8 neutrons/s, is on a civilian-deployment trajectory for the late 2020s and early 2030s. The same compression physics, applied with chemical explosives rather than pneumatic pistons, is the closest-to-weaponizable pure-fusion approach in the open literature. The IEER's 1998 estimate of a 1-tonne-class deliverable pure-fusion device is not far-future speculation; it is consistent with the LM26 results.

8.1 Outlook for 2026-2045

Looking forward to the 2026-2045 horizon that frames the TRL roadmap in Figure 6.2, the most likely developments are: (a) NIF yields continue to increase as the program refines target design, with multi-MJ yields becoming routine by the late 2020s; (b) LM26 reaches 1 keV and then 10 keV plasma temperatures in successive campaigns, with the first fusion-relevant compression milestone possible by 2027-2028; (c) BASE and ALPHA continue incremental progress, with antihydrogen accumulation reaching 10^5-10^6 atoms by 2028 and transportable traps becoming routine; (d) fast ignition moves toward TRL 7 as petawatt-laser facilities (BELLA at LBNL, LFEX at Osaka, Shenguang-IV in China) reach the necessary intensities and collimation uniformity; (e) no pure-fusion weapon is openly developed or tested by any state, but the industrial base for one continues to mature.

For policy analysts, the implication is that the 2025-2035 decade is the window in which threshold-yield declarations, no-first-pure-fusion-weapon declarations, and the safeguarding of large antimatter-production facilities should be put in place. The CTBT's Article I scope already covers the relevant activities, in the open-literature reading; what is needed is not new treaty law but political will to apply the existing scope to the new facilities. The technical community has done its part by demonstrating that fissile-free ignition is physically possible; the policy community now has to decide what to do with that demonstration.

8.2 Recommendations for further open-literature research

For researchers wishing to extend this synthesis, the highest-value open-literature directions are: (1) systematic comparison of the published antiproton-count estimates across the Gsponer-Hurni, Perkins-Orth-Tabak, Shmatov, and Azizi papers, with attention to the assumptions on dE/dx, ε, and κ; (2) monitoring of LM26 results as the program advances from 0.72 keV to 1 keV and 10 keV; (3) tracking of BASE and ALPHA publications for storage-time and antihydrogen-count milestones; (4) analysis of the AMBER cross-section measurements as the first new antiproton-production data in three decades; (5) policy analysis of the CTBT Article I scope question raised by the Bulletin article, including engagement with the NPT Review Conference process on the civilian-fusion-research non-proliferation dimension.

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